Rohto Pharmaceutical (4527) research system · how one daily run flows

Fukuhara Seminar · Hugo Hayashi · skill flow v4 (2026-11-13). Read top to bottom: the order the system runs (middle), what I do and decide at each point (left), and the outside data each stage pulls (right). "LATEST" lines quote the newest files. Data site: **h-hayashi-ctrl.github.io/rohto-dashboard**.

→ pipeline order · ⇢ I act · YOU DECIDE stops for a human · AUTO GATE a rule decides · ① – ⑦ stages · A1 fundamental A2 alternative data A3 competitor A4 debate and decision

ME · INPUTS AND DECISIONS · **THE SKILLS** · IN RUN ORDER · **OUTSIDE DATA** · PULLED LIVE

Before a session
you
Sync from the server the day before the upload deadline and check the Kuma monitor is green, so every number is from the same run.

0 The clock starts a run — no command needed · home server

Daily run on the home server
`fukuhara-daily.timer · daily_fetch.sh → daily_agents.sh`
systemd user timer on zanet (over Tailscale), every day 18:30 JST: fetch, Agents 1 → 2 → 3, health check, Agent 4, snapshot. Runs whether the laptop is on or not. Logs in output/logs/.
LATEST last daily run 2026-11-13 (synced) · every day 18:30 JST

Uptime Kuma heartbeat
`daily_agents.sh → Push monitor`
AUTO GATE
Sends up, or down with the failed step; no push for ~25 h also turns it down. The message carries today's position and re-debate status.
LATEST push URL set · down on a failed step or after ~25 h with no run

Cycle on request
`skill: weekly-research-run · sync_from_server.py`
Ask Claude Code to "run this week" on the laptop: it first pulls the server's data and outputs, then runs the same order.

Session calendar
`syllabus`
Uploads are due one day before each session; the cycle is rerun the day before.
LATEST Session 5 was 13 Nov; next date not set

runs ① → ⑦

1 Collect public data — every day · always

A1 Market data
`fetch_market.py · market-data`
Price, consensus EPS, analyst targets, 6 years of monthly prices. Secondary source: used for checks and comparison only.
LATEST price ¥2,629 (2026-11-11) · consensus FY27 EPS ¥159.4 (7 analysts)

A1 Filings
`fetch_edinet.py · filing-extraction`
5 years from the annual reports (XBRL); every number keeps its document, page and tag. Rerun when the fiscal year changes.
LATEST 85 values extracted · checked against the csv

A2 Alternative data
`fetch_altdata.py · alt-data-fetch`
Brand and symptom searches, share of search in 4 categories, Tokyo UV, JNTO visitors, cosmetics exports to China, Rakuten top-30. Keeps the last good data when a source fails.
LATEST 12 search terms · 2026-11-13

A3 Competitor disclosures
`fetch_competitor_news.py · competitor-disclosures`
TDnet filings of Rohto + 7 peers for 180 days; reads up/down from the title, or the PDF while it is still online (~30 days).
LATEST 62 filings from 8 companies · 2026-11-13

A2 Regional data
`fetch_regional.py`
Google Trends by prefecture for the same terms, and Rohto's share of search per prefecture. By hand, about monthly (prefecture gaps move slowly). e-Stat layers (drugstores, symptoms, medicine spending) wait for an API key.
LATEST 12 terms + 4 share sets × 47 prefectures · 2026-11-13 · e-Stat waiting for ESTAT_APP_ID

Public sources
robots and terms respected
EDINET API · Yahoo Finance (yfinance) · Google Trends · Open-Meteo · JNTO · UN Comtrade xlsx · Rakuten Ichiba API · TDnet list via yanoshin WebAPI (no key) · e-Stat API (key pending)

every run

2 Check the run — always

Fix what is flagged
you
Check each gap in the original filing; correct the csv myself (student-corrected + note). The agent never overwrites a value.
LATEST 6 gaps to check in the filings

A1 Data integrity
`baseline_check.py`
Missing fields, mixed units, conflicting values, unverified and secondary sources → error / warning / info. Any error fails the health check.
LATEST error 0 · warning 28 · info 14

A1 Cross-source check
`baseline_check.py`
Filings vs. yfinance; a gap above 0.5% becomes SOURCE_MISMATCH. The agent never decides which one is right.
LATEST 6 gaps to check in the filings

Health check
`check_health.py`
AUTO GATE
Fails if a forecast source is missing or older than 8 days, Rohto's own filings are missing, any agent has an error, or outputs come from different runs. On fail Agent 4 only checks thresholds, never debates.
LATEST OK · 13 ok / 1 warn / 0 fail · 2026-11-13 04:31

store counts given? (otherwise skip to ④)

3 Field layer — only with my counts

Collect a store visit
you
Count facings by the protocol; shelf photo, no faces. Never enter a count I did not see.
LATEST no store visits yet

A3 Observation sheet
`field/test_observations.csv · protocol.md`
Course schema: facings, price, promotion, stock-out and special conditions; one row per SKU per visit, same store and same time slot.
LATEST no store visits yet

A3 Shelf share + stock-outs
`competitor_check.py`
AUTO GATE
Rohto facings ÷ all facings, change over ≥3 visits of the same store × category; stock-out rate vs. peer national brands. Scored only when the rule passes.
LATEST waiting for ≥3 visits per store × category

A3 Field data quality
`competitor_check.py`
Flags bad facing counts, too few visits and sale or renovation days. Nothing is filled in.

daily: Agents 1 and 2

4 Evidence → forecast → value — daily · Agents 1 · 2

Read and decide
you
Every placeholder is tagged source: agent. Accept it or replace it in the assumptions file with source: student and a reason, then rerun.
LATEST 44 placeholders still tagged source: agent

A1 Agent 1 · Forecast
`baseline_check.py · baseline-check`
Company guidance + growth and margin assumptions → FY27–28 EPS and next-12-month EPS; gap to consensus explained, not copied.
LATEST NTM EPS ¥162.1 · implied P/E 19.0×

A1 Agent 1 · Own multiples
`baseline_check.py`
P/E and EV/EBITDA on next-12-month numbers; target P/E checked against the 6-year P/E band.
LATEST ¥2,901 (P/E ¥2,918 · EV/EBITDA ¥2,883)

A1 Agent 1 · Peer valuation
`peer_check.py · peer-valuation`
Median P/E and EV/EBITDA of 6 peers, ROE–P/B regression, P/E decomposition; blended with own multiples by the peer weight.
LATEST peers ¥3,250 × 0.5 → Agent 1 ¥3,076

A2 Agent 2 · Alt signals
`alt_check.py · alt-signals`
AUTO GATE
Fiscal-year-to-date YoY × elasticity → Japan revenue → EPS → target price (a proposal). Confidence +0.05 when signals agree, −0.05 when they conflict.
LATEST brand +8.5% · symptoms -4.8% → Japan revenue +0.6% · TP +¥14 · confidence 0.30 → 0.25

Agents 1 and 2 done

5 Agent 3 · Competitor intelligence — daily · Session 4

Fill what the agent cannot read
you
YOU DECIDE
Enter the direction of each unknown revision; confirm whether Rohto's growth includes the Thann Oryza acquisition.
LATEST 3 revision without a direction (Rohto 2026-11-10)

A3 Event classes
`fetch_competitor_news.py`
12 classes in priority order (TOB, guidance, M&A, product and price, strategy, impairment, restructuring …). My own entries are kept on every refetch.
LATEST 6 peer events flagged (restructuring, M&A) in 180 days

A3 Guidance direction
`data/competitor/disclosures.csv`
YOU DECIDE
When neither the title nor the PDF says up or down, the row stays unknown and is not scored until I read the filing.
LATEST 3 revision without a direction (Rohto 2026-11-10)

A3 Operating scorecard
`competitor_check.py · competitor-check`
Revenue growth and operating margin vs. the peer median (company forecasts); flags fiscal-year mismatch and acquisition-driven growth.
LATEST growth Rohto 8.3% vs median 4.0% (+1) · margin Rohto 12.1% vs median 9.7% (+1)

A3 Share of search
`competitor_check.py`
Rohto's share in 4 categories vs. last year. Same data as Agent 2, so it moves the multiple only, never EPS (no double count).
LATEST +0.98pt across 4 categories (+0)

A3 Ad & promotion spend
`fetch_ad_spend.py · competitor-ad-spend`
Advertising + sales promotion from the annual reports (XBRL) of Rohto and 7 peers; promotion efficiency = sales growth − A&P growth vs. the peer median.
LATEST A&P 18.2% of sales · sales +11.4% vs A&P +8.9% → +4.7pt vs peers (+1)

A3 3C × 4P map
`competitor_check.py (framework)`
Every piece of evidence filed under Customer / Competitor / Company and Product / Price / Place / Promotion, marked scored, reference or waiting. Price is shown, never scored.
LATEST Customer scored · Competitor ref · Company scored · Product ref · Price ref · Place ref · Promotion scored

A3 Peer-multiple premium
`competitor_check.py`
AUTO GATE
+1 / 0 / −1 per item × 2%, capped at ±6%, on Agent 1's peer valuation. Confidence +0.05 when ≥3 items agree; never raised without store data.
LATEST score +3 → +6% → TP +¥98 (¥3,076 → ¥3,174) · confidence 0.25 → 0.25

Competitor filings
no login, no personal data
TDnet (8 companies) · TDnet PDFs read with pdftotext · EDINET annual reports (A&P) · Google Trends share of search · Rakuten ranking (saved)

all three agents done · health check OK

6 Agent 4 · Debate and decision — checks daily · debates on a trigger · new · Session 5

Read the debate
you
YOU DECIDE
Read debate_log.md; accept or override the moderator's number and write why. Agent numbers stay labelled as agent numbers.

A4 Re-debate triggers
`decision_rules.py · decision-check (daily)`
AUTO GATE
Price ±5%, consensus EPS ±3%, a Rohto guidance / M&A / TOB filing, 14 days, or a flipped position → redebate. Checked every day after Agents 1–3 and the health check.
LATEST 2026-11-13 · hold (none) · now overweight

A4 Agent 4 · Debate
`run_debate_agent.py --if-needed · debate.py · valuation-debate`
AUTO GATE
Runs only on redebate (Claude Code on the server) or when I ask, and only if the health check passes. Opening → rebuttal → final; a moderator weights the three.
LATEST 2026-11-13 · 3 agents · agreed ¥3,103

Final target price
`debate_log.md · final_valuation.json`
YOU DECIDE
The agents propose; I set the final target price and confidence for the upload, with reasons.
LATEST proposed ¥3,103 · confidence 0.25 · overweight

A4 Position
`decision_rules.py · decision-check`
AUTO GATE
Expected return − 6% benchmark = alpha; × confidence → overweight / neutral / underweight at ±0.5%. Back-solves the price, target and confidence that would flip the call.
LATEST Agent 1: return 18.9% · alpha 12.9% × 0.25 = 3.2% → overweight

approved

7 Record and publish — cards and snapshot daily · site when I run it

Send
you
The server sends the agent cards and the snapshot every day; the site and the skill pack go out only when I sync and run them. Keys only in .env.

Class dashboard
`dashboard_connector.py · dashboard-report`
Each of the 4 agents sends to its own card every day; anything unsent is queued and resent next run.
LATEST 4 agent cards

Daily snapshot
`snapshot.py → data/history/snapshots.csv`
One row a day: price, targets, position, flip price, signals, health. Feeds the site's "Since last week" table and its rule-written conclusion.
LATEST 13 days recorded (2026-10-30 → 2026-11-13; 1 from daily runs)

Data site
`sync_from_server.py → publish_site.sh`
GitHub Pages charts rebuilt from the agent outputs and raw data, with Since last week and a prefecture map (Regions); no new calculation.
LATEST data rebuilt 2026-11-13 11:48

Skill pack
`build_skill_pack.py`
All skills, scripts, data and this flow in one zip for my partner; keys are never included.
LATEST rohto-agent-skills_2026-11-13.zip · 0.5 MB

Where it goes
published
Seminar dashboard (agent cards, monitors, activity) · h-hayashi-ctrl.github.io/rohto-dashboard · Slack (skill pack)

What v4 changes vs v3 (30 Oct)

- **Daily, on a home server:** the whole run moved from a Monday laptop timer to zanet at 18:30 JST every day; the laptop only syncs and publishes.
- **Agent 4 split out:** debate and decision are their own agent (2 skills, own dashboard card); it checks the thresholds daily and re-debates only on a trigger.
- **Explicit decision thresholds:** the ±0.5% band and the price, target and confidence that would flip the call are written down and checked.
- **Monitored:** an Uptime Kuma push monitor goes down on any failed step or a missed day.
- **Since last week:** a daily snapshot lets the site compare with 7 days ago and write the conclusion by rule.
- **Regions:** prefecture map of brand and symptom searches and Rohto's share of search, with correlations.

Rules every stage follows

- The agents extract, check, calculate and challenge; I decide.
- Missing data stays missing; nothing is imputed; no store data is invented.
- Every input is tagged agent / company / student, with a reason.
- Every warning is shown; confidence is never raised on weak evidence.
- Keys only in .env; never in code, logs or the skill pack.

Course work, not investment advice. Generated by scripts/build_skill_tree.py from the latest output files.